2.5 Measures of Time

The teacher asks students to answer the following questions:

- How long do you take to run 100 metres?
- How long do you take to walk one kilometre?
- What is the time taken for a cup of rice to be cooked?
- What is the cultivation period of groundnut?

These questions will help us to find the importance of time in our day-to-day life.
Now let us discuss the development of measures of time.

The procedure of measuring time has undergone several changes from ancient period. Initially, a stick in the sand was used to measure time by measuring the length of the shadow of that stick. Then horizontal and vertical plates were used as sundials to measure time between sunrise and sunset, that is in the day time. Firing of knotted ropes was used to measure time in the darkness. The approximate time taken for the fire to travel from one knot to other formed the part of night. In later days, a day was divided into 24 equal parts (hours) of which 12 hours were for daytime and 12 hours were for night time approximately. Time taken by the Earth to complete one full rotation around the Sun is known as the **Solar Year**. It was divided into 12 equal parts which is known as the **Solar month**. The duration between two full moons is known as the lunar month and 12 lunar months are known as lunar year. But we follow solar year and month.

Various clocks had been designed and used to measure time from different parts of world, like water clock, sun dial candle clock, sand clock, rope clock, etc.... Have you seen those clocks? Look at the clocks shown below.

			
Water clock	**Sun dial**	**Candle Clock**	**Sand Clock**

Study of devices that are measuring the time is called **'HOROLOGY'**.

Nowadays, we use pendulum clock,digital clock, quartz clock, atomic clock to find time accurately.



Our Tamil people were experts in the Astronomical science. The Tholkappiam deals the pozhuthu (time). They divide a day into six major divisions, together called **"Sirupozhuthu (சிறுபொழுது)"** . A year into six major divisons, together called **"Perumpozhuthu (பெரும்பொழுது)"**

1 Nazhigai = 24 min; 1 hour = 2.5 nazhigai = 1 Orai; 1 day = 24 hours = 60 nazhigai

[Tamil people used the device " kuri neer kannel" to measure night times.....]

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